Simulation of ICFO – Collserola free-space link

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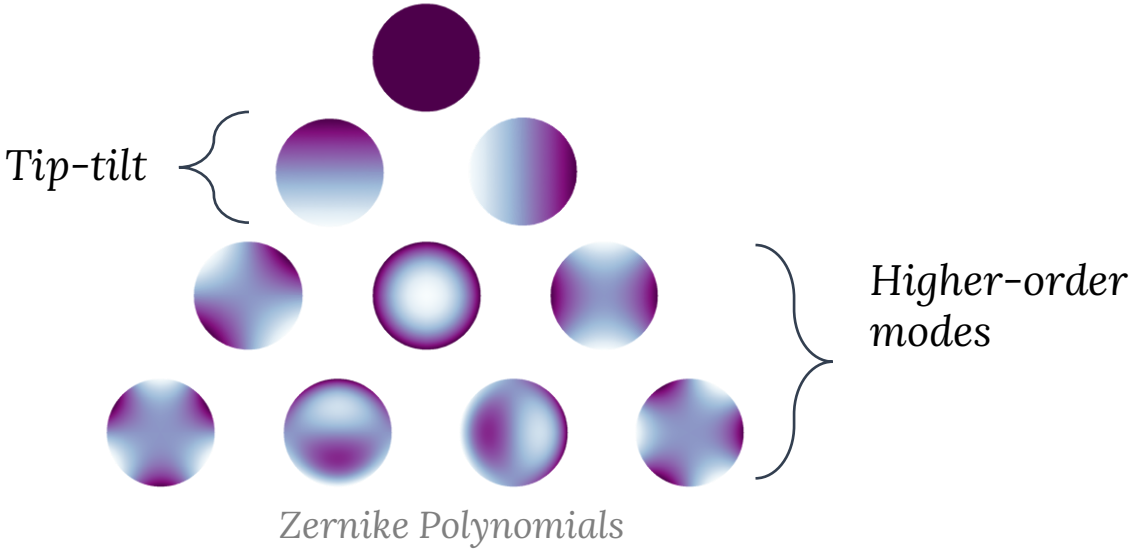
Photo: Felix König

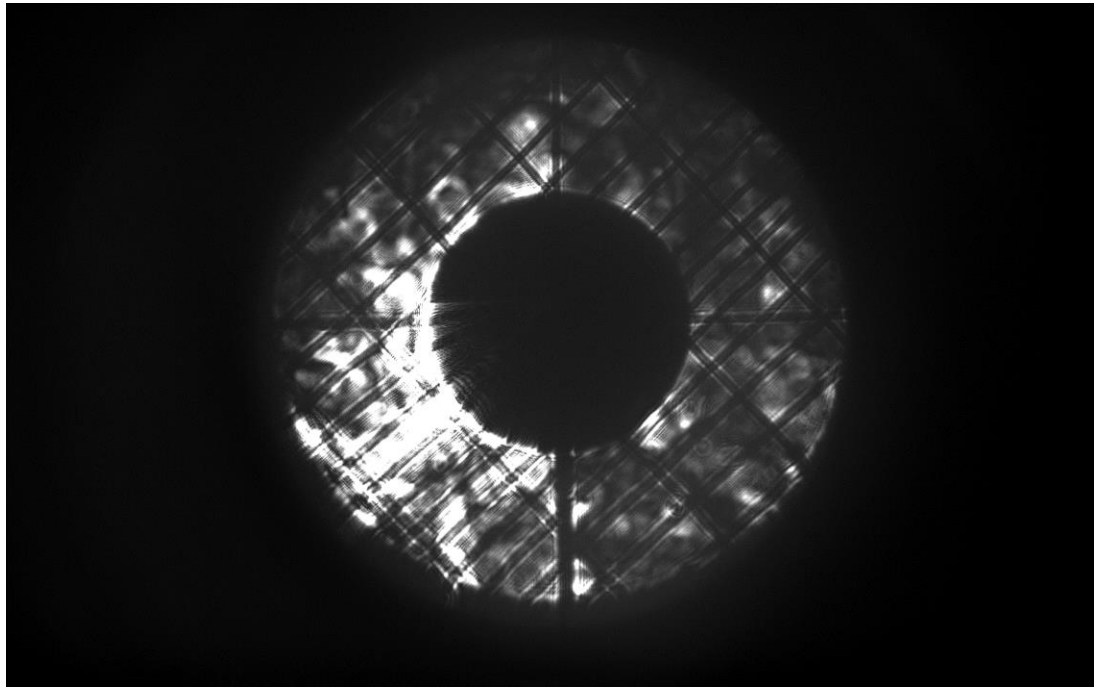
Free-Space QKD

- Polarization-based BB84 at 1550 nm
- Daylight conditions
- Challenges
  - **Background noise**
    - Spectral filtering
    - Temporal filtering
    - Spatial filtering with single-mode fiber (SMF)
  - **Atmospheric Turbulence**
    - Adaptive optics (AO) techniques
  - **Environmental conditions**
  - **Synchronization**
    - Beacons
    - GPS signal

Atmospheric turbulence

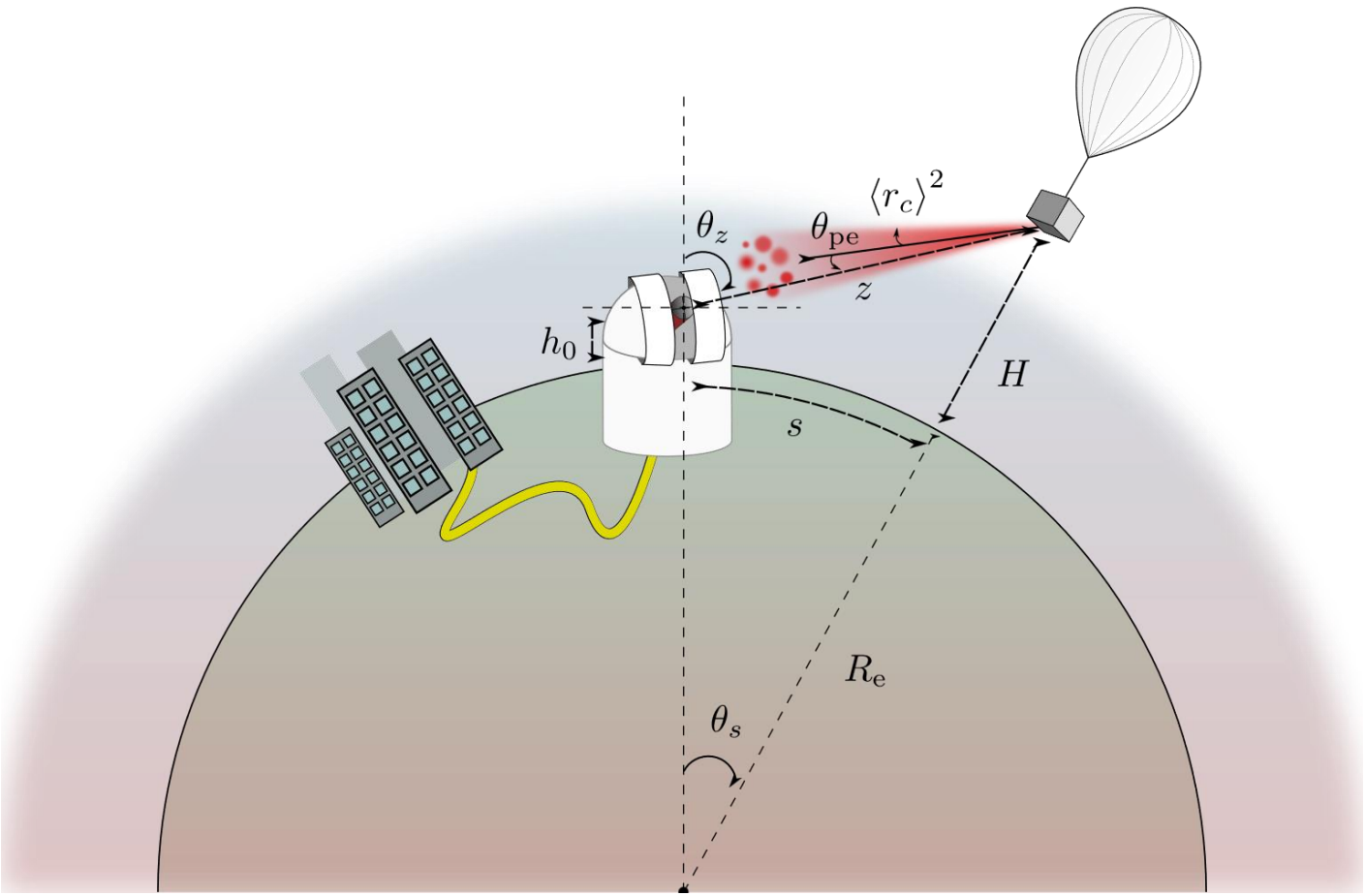
- Beam wandering
- Beam broadening
- Beam deformation
  - Scintillation: Rytov variance σ_R^2
  - Phase fluctuations: Coherence width r_0





Channel simulation

- Models for downlink, horizontal and uplink for high-altitude platform networks [1]
- Applicable for large zenith angles and strong scintillation conditions
- Used with event-based network simulator (Netsquid) to determine feasibility of different protocols
- Parameter exploration
- Open-source Python code [2]



[1] I. Karakosta-Amarantidou, R. Yehia and M. Schiavon, arXiv:2412.03356 (2024).
[2] I. Karakosta-Amarantidou and R. Yehia. *balloon_qnet*. GitHub repository. 2024. url: https://github.com/RajaYehia/balloon_qnet.

Free-space loss model

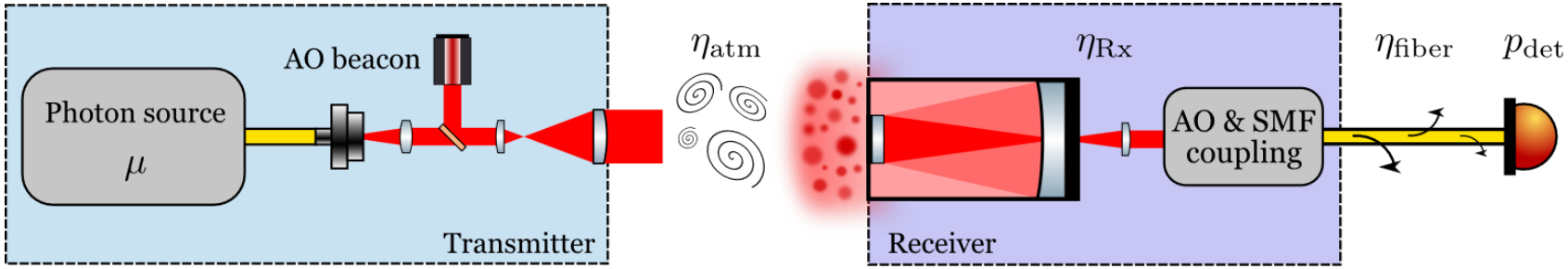
- Model inspired from [3]
- Downlink channel example
- Horizontal channels with passive receivers
- Atmospheric absorption from LOWTRAN [4] (Python wrapper [5])
- Statistical distributions for fiber-coupling efficiency [6] and collection efficiency [7]
- Uplink modelling based on reciprocity principle [8]

Atmospheric absorption

Collection efficiency

Coupling efficiency

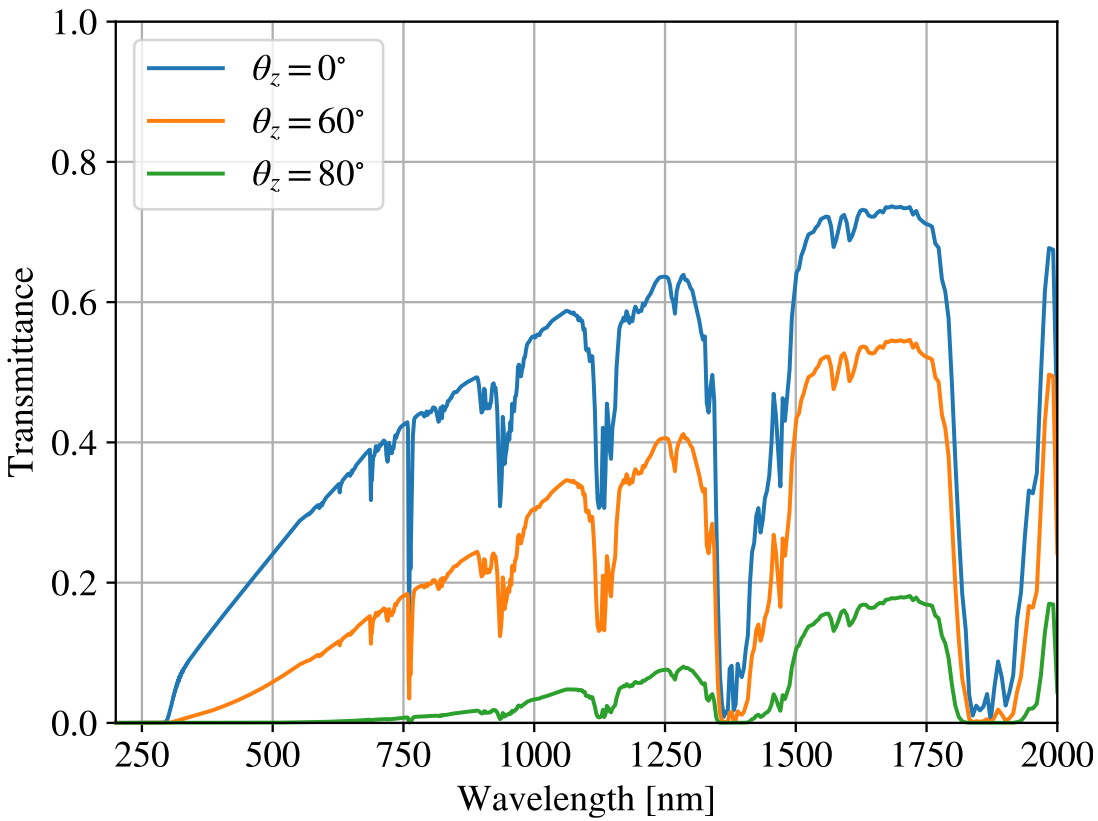
$$\eta_{\text{free-space}} = \eta_{\text{atm}} \cdot \eta_{\text{Rx}} = \eta_{\text{atm}} \cdot \eta_{\text{D}_{\text{Rx}}} \cdot \eta_{\text{SMF}}$$



[3] A. Scriminich et al., *Quantum Science and Technology* 7.4 (2022): 045029.
[4] F. Kneizys, E. Shettle, L. Abreu, J. Chetwynd, and G. Anderson, Air Force Geophysics Laboratory AFB MA (1988).
[5] F. Picciariello, *lowtran-piccia*. GitHub repository. 2024. url: https://github.com/francescopiccia/lowtran-piccia.
[6] L. Canuet, N. Vedrenne, J.-M. Conan, C. Petit, G. Artaud, A. Rissons, and J. Lacan, JOSA A 35, 148 (2018).
[7] D Vasylyev, W Vogel, and AA Semenov, Physical Review A 97.6 (2018), p. 063852.
[8] C. Robert, J. M. Conan, and P. Wolf, Physical Review A 93.3 (2016): 033860.

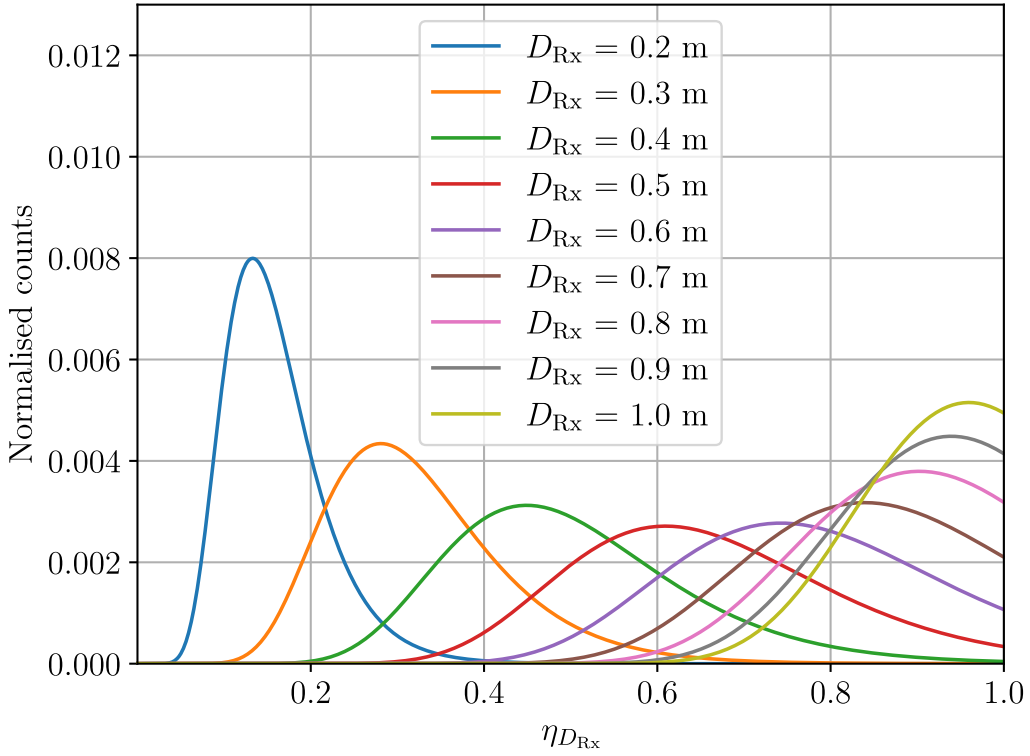
Atmospheric transmittance

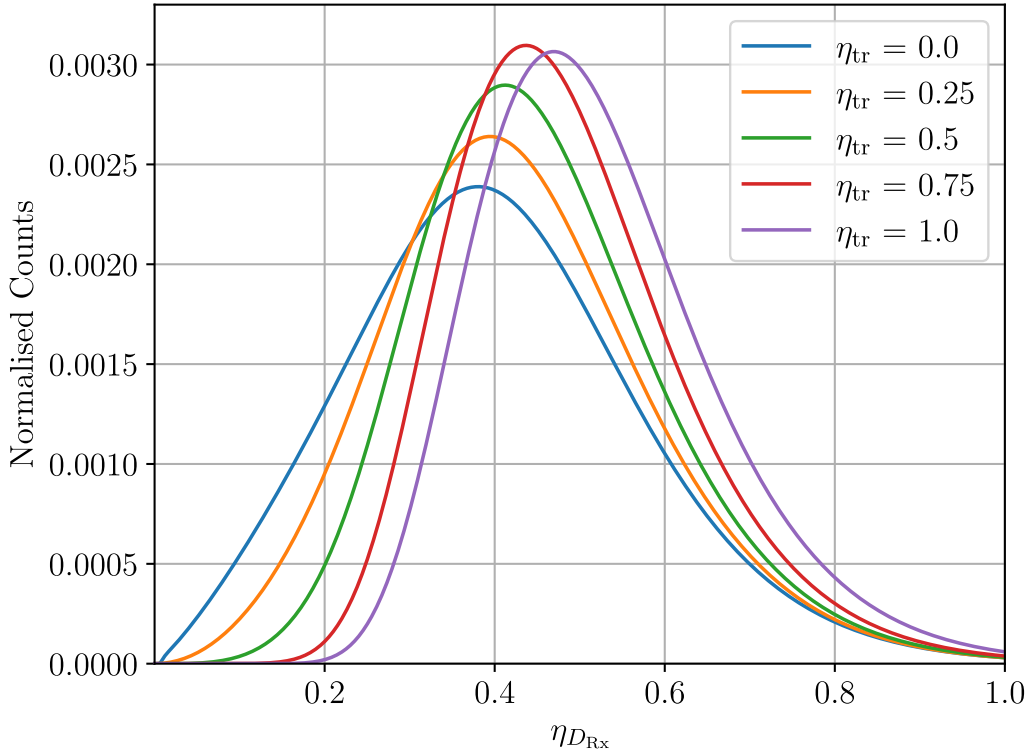
- Parameters
  - **Wavelength**
  - **Zenith angle**
  - **Altitude of terminals**
  - **Atmospheric model** (tropical atmosphere, midlatitude winter, subarctic winter…)
  - **Visibility**
  - Season
  - Cloud/rain conditions
  - …



Collection efficiency

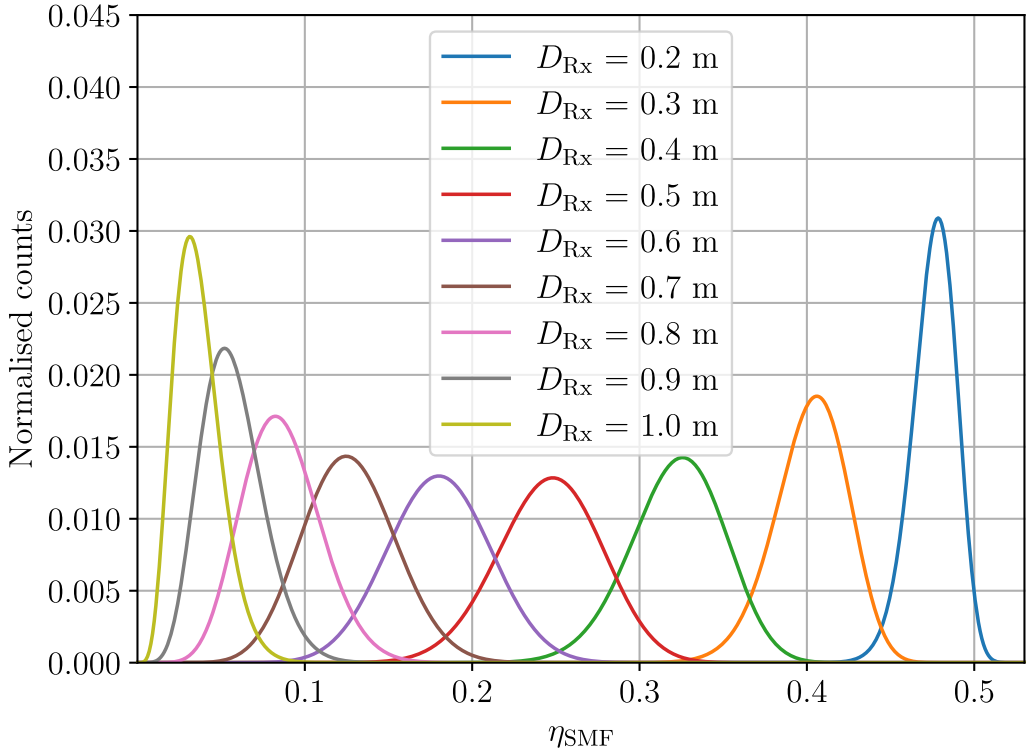
Condition: small beam wandering

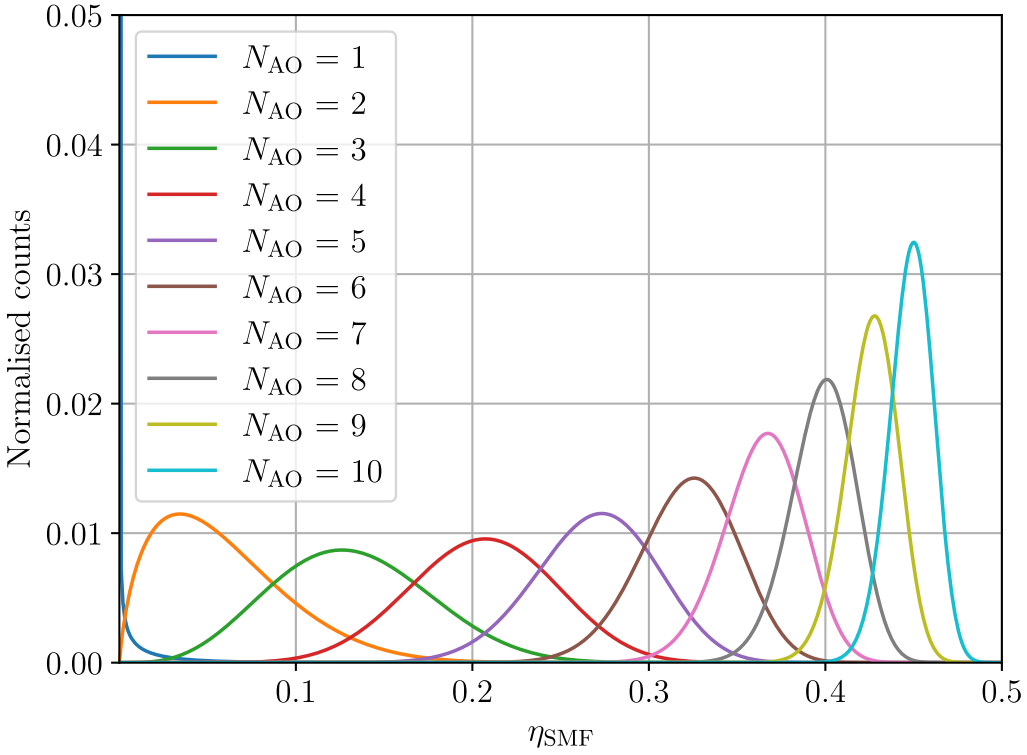




Coupling efficiency

Condition: small residual phase

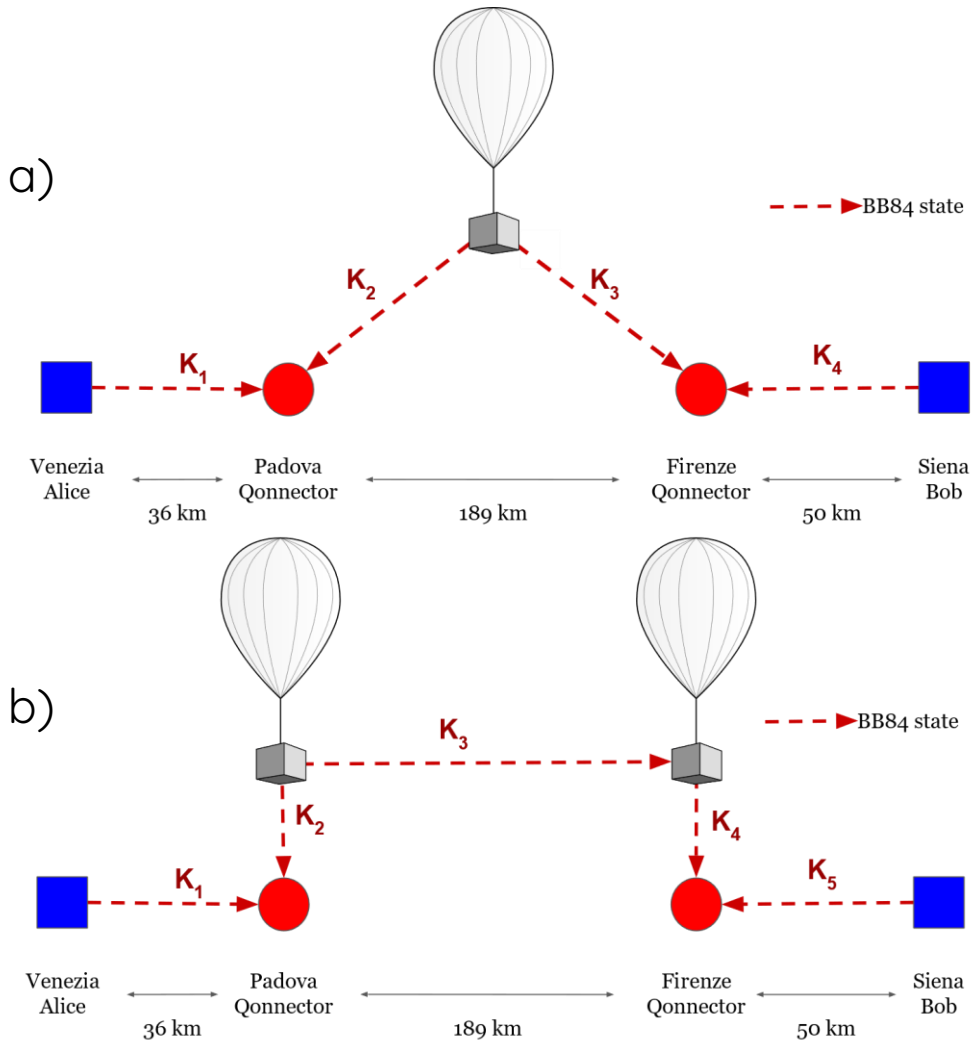




Trusted balloon nodes

- Each node performs BB84 with its neighbor
- Upper bound for secret key rate
- Performance determined by minimum rate achievable
- Two balloons setup more efficient, but requires more hardware

Link	Secret key rate (kbps)
Slant link Balloon - > Qonnector	**2.21 ± 0.144**
Vertical downlink Balloon - > Qonnector	112.01 ± 0.735
Horizontal link Balloon - > Balloon	**24.65 ± 0.451**
Alice - > Qonnector	70.71 ± 0.685
Bob - > Qonnector	39.85 ± 0.371

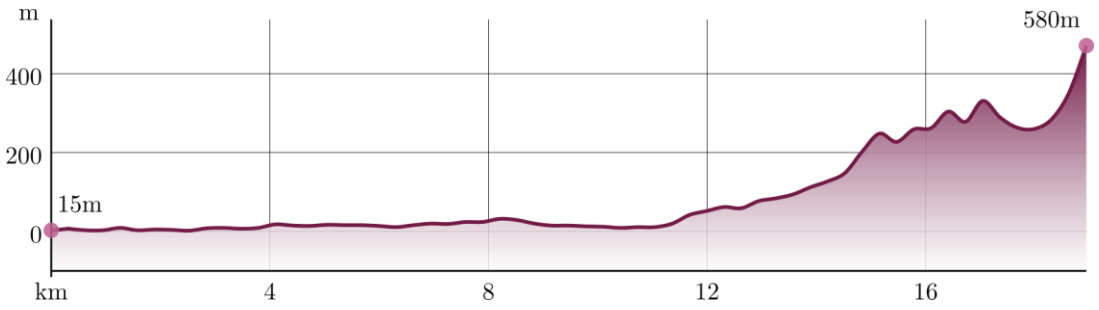


Link parameters

Parameter	Value
Transmitter altitude	~15 m
Receiver altitude	~580 m
Channel length	18.97 km
Zenith angle	88.34°
Waist radius at Tx	30 mm
Average transverse wind speed	10 m/s
Wavelength	810 nm
Pointing error	0 urad
Tracking efficiency	0
Aerosol model	Rural - visibility 23 km
Rx obstruction ratio	**0.47**

- For ICFO – Collserola link, only collection efficiency and atmospheric absorption
- Obstruction ratio is not currently taken into account at collection efficiency

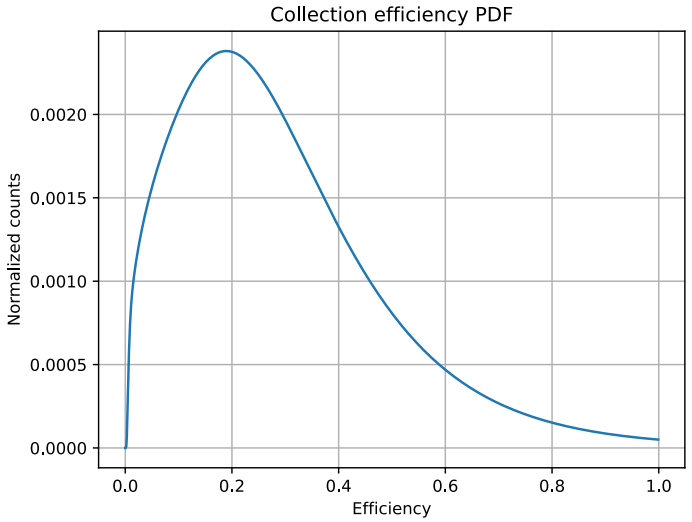
$$\sigma^2_{\text{wander}} = \left[\left(z\theta_{\text{pe}}\right)^2 + \langle r_c^2 \rangle\right] \cdot \left(1 - \eta_{\text{tr}}\right)$$

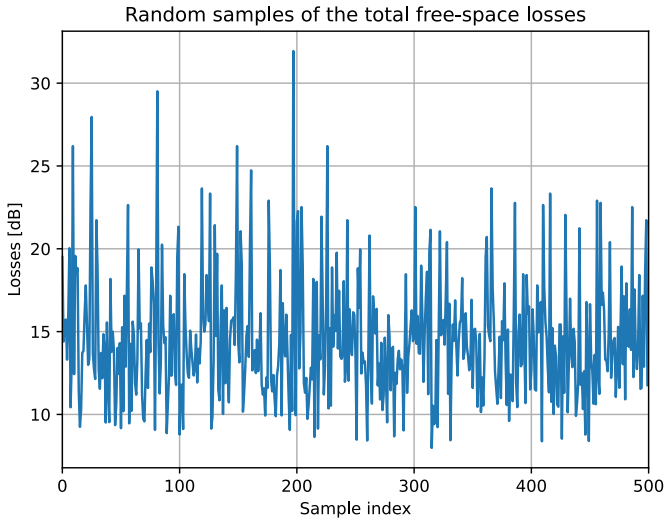


Profile generated from [Printmaps.net]

Results

Parameter	Value
Atmospheric transmittance	0.16
Rytov variance	9.45
Long-term beam size at Rx	76.4 cm
Average losses [dB]	14.5 dB





Thanks to my colleagues:
Matteo Schiavon
Raja Yehia

Thank you for your attention!
Questions?

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